

1. Overview

Ultimate Goal:

- Create a system that can map out differences between prediction markets and stock movement through Kalshi, Polymarket and the general stock market

Key Features:

- **News Aggregator:** poll news api based on specific filters (news source, time, topic, ...) and outputs a list of relevant news
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 - **Filters:** Source IDs, Source category, Country, Language, Keyword query, Exact-phrase and boolean terms, Time window, Sort order, Domain allowlist or exclusion list, **Maximum results per poll**, **Poll interval**, and title similarity, Minimum article length or require full text, Exclude opinion, sponsored, or live-blog content, Ticker or company name match
- **Twitter Aggregator:** poll twitter api based on specific filters (hashtags, user, ...)
 - **Filters:** Specific accounts, Exclude accounts, Hashtags, Cashtags, Keyword and phrase queries, Boolean operators and exclusions, Language, Contains links, Minimum engagement, Account verification or account age, Time window, Geography, Poll interval and rate-limit budget, Deduplication of retweets and quotes
- **Jev Classifier:** classifies news/twitter events based on specific questions (can be user defined or just positive/negative)
- **Kalshi / Polymarket Fetcher:** Allow a user to look through certain prediction markets to overlay on top of stock data
- **Company Network:** relational graph of companies (how would we create this graph? Thoughts: by industry, by supply chain, by production, by distribution) (sector, subindustry, business model, market cap band, listing venue, competitors, partner, dependencies)
 - **Filters:** Minimum average daily dollar volume, Market cap floor, Single exchange or venue restriction, Minimum days of price history, Active status only, Minimum edge weight, Relationship type, Source confidence, Edge recency, Reciprocal confirmation, Hub cap, Minimum degree, Minimum co-mention count, Lookback window length, Rolling rebuild schedule, Minimum cluster size, Maximum cluster size, Stability across seeds or shifted windows, Coherence
- **Trending Cards:** Provide the user with new and recent summarized context based on aggregators and prediction market fetcher to explain market trends

- **Display Charting:** Allow the display of Stock charts overlapped with prediction market charts to view correlation between related markets
- **News Graphing ? (Change name) :** Indicate where certain pieces of news data corresponds on top of the displayed charts (Example: little dot on the graph indicating a certain news event)
- Group related data sources
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Option 1

Prediction Markets as a Financial Signal

Prediction markets like Polymarket and Kalshi price real-world events in real time, from Fed decisions and inflation prints to tariffs and elections. Build something that turns this data into insight for investors, researchers, or analysts.

Your project could

- Detect and explain sudden market moves
- Connect markets to companies they affect
- Test if markets lead or lag stock prices
- Build a monitoring tool or AI assistant

Data and tools

Polymarket, Kalshi's public API, and SEC EDGAR. Live or historical data both work, and pairing market data with filings or news could be helpful.

Option 2 (General Track)

AI for Finance

Use AI to build something that helps traders and analysts research companies, analyze markets, and make decisions.

Your project could

- Flag companies worth watching today
- Gauge company sentiment using public discourse and news
- Try any creative use of AI in finance!

Requirements

- A clearly defined user and problem
- Public datasets only
- Some form of evaluation, like a backtest